Shivam Singh

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EDUCATION

B.Tech - Electronics and Communication Engineering Uttar Pradesh, India | 2017-2021
Raj Kumar Goel Institute Of Technology, Ghaziabad CGPA: 8.07/10

MS - Control and Optimization Chennai, India | 2022
Indian Institute of Technology Madras CGPA: 8.5/10

MS - ECE Hyderabad, India | 2023-2025
International Institute of Information Technology, Hyderabad CGPA: 8.33/10

WORK EXPERIENCE

ALSOENERGY – INDIA | Technical Support Engineer April 2021 | Dec 2021

- As a Technical Support Engineer at AlsoEnergy, I provided expert assistance and troubleshooting for clean energy management systems, specifically focusing on the software platforms like powertrack.
- I collaborated closely with clients, including commercial solar developers and energy managers, to diagnose and resolve technical issues related to system performance, data visualization, and integration with third-party hardware.
- My role involved conducting in-depth analysis of client-reported issues, guiding users through step-by-step diagnostic procedures, and delivering immediate solutions for common challenges.

RESEARCH EXPERIENCE

AMD INDIA PRIVATE LIMITED | Research Fellow AI Models & Agents Team | Jun 2025 - Jun 2026
Collaborators: Saptarshi Majumder, Pratik Prabhanjan Brahma, Zicheng Liu, Emad Barsoum

- Pause and Think: Video-Grounded Assistive Action Suggestion: *2025*
  - Built *Pause-and-Think-T*, a reasoning-centric training corpus of ∼10k video QA pairs, and *Pause-and-Think-B*, a 300-sample benchmark for grounded contextual understanding and goal planning, spanning Epic-Kitchens, Assembly101, and Ego4D (egocentric and exocentric views).
  - Designed the three-stage data pipeline: annotation refinement with gpt-oss-120b, goal-oriented clip segmentation (5–20s), and structured `<think>`/`<answer>` QA generation via Qwen3-VL-235B with majority voting and self-consistency verification.
  - Fine-tuned a compact 4B-parameter VLM (using LLaMA-Factory on 8×AMD Instinct MI325 GPUs, served with vLLM) reaching 58.0% on the benchmark – matching Qwen3-VL-235B (58.9%, 59× larger) and GPT-5.2 on scene understanding, while surpassing GPT-4o; demonstrated OOD gains on EgoThink and TempCompass without benchmark-specific training.
  - Deployed the fine-tuned model on an AMD RyzenTM AI board, enabling real-time on-device inference for edge use cases.
  - **My contribution:** As first author, led the dataset design, pipeline implementation, fine-tuning, evaluation, and paper writing.
  - Paper Link : https://arxiv.org/abs/2606.00616
- Spatial Video Understanding from Raw RGB Videos: *2026 - ongoing*
  - Developed a pipeline that performs spatial understanding, object tracking, and goal planning directly from raw, real-world monocular RGB videos.
  - Combined multiple vision models (Depth-Anything-3, YOLO-World, YOLO26) with VLMs into a single pipeline that takes a raw monocular RGB video as input and produces a 3D map of all object locations along with the user's pose.
  - Used this pipeline to generate a clean and robust QA dataset for object tracking, spatial understanding, and planning.

- Training an end-to-end model that, given a raw monocular RGB video, reasons over the spatial scene temporally using built-in memory and 3D understanding to answer user questions in natural language.

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY - HYDERABAD | RESEARCH STUDENT
Robotics Research Centre, Hyderabad | Jan 2023 - May 2025
Direct Supervisor: Prof. Madhava Krishna
Collaborators: Dr. Mohan Sridharan, Dr. Krishna Murthy Jatavallabhula, Dr. Brojeshwar Bhowmick

- Task Anticipation using LLMs: *2023*
  - Developed an intelligent household agent that learns task execution patterns by extracting user behavior and preferences using various Large Language Models (LLMs) and designing task and action representations in Planning Domain Definition Language (PDDL).
  - My contribution: As a co-author & project lead, I played a key role in generating a diverse state and action space in PDDL and designing LLM prompts. I also contributed significantly to the writing process.
  - Published in IEEE ICRA 2024.
  - Project Website : https://raraghavarora.github.io/ahsoka/
  - Paper Link : https://arxiv.org/pdf/2502.02066
- Task Anticipation for Multi-Agent Systems: *2024*
  - This work extends LLM-based task anticipation to multi-agent settings, where robotic agents anticipate tasks within their domain, adapt to unexpected changes in human behavior, and account for uncertainties in the environment to collaborate effectively in human-robot scenarios.
  - My contribution: As project lead, I designed the domain description and formulated the framework that adapts to dynamic human behavior and environmental changes.
  - Accepted at the ICRA 2024 and RSS 2024 workshops.
  - Project Website : https://dataplan-hrc.github.io/
  - Paper Link : https://arxiv.org/pdf/2404.03587
- Combining LLM with Knowledge Graphs for task decomposition and knowledge refinement: *2025*
  - Designed a framework combining Large Language Models (LLMs), Knowledge Graphs (KGs), and Human-in-the-Loop (HITL) approaches to enable embodied agents to handle unseen tasks and adapt to new scenarios.
  - My contribution: As project lead, I designed and implemented the system framework and conducted experiments with various LLMs (GPTs, LLaMA, Mistral, Gemini).
  - Accepted in IEEE ICRA 2025.
  - Project Website : https://sssshivvvv.github.io/adaptbot/
  - Paper Link : https://arxiv.org/pdf/2502.02067
- Hybrid Planning with LLMs and Probabilistic Models in Multi-Agent Settings: *2025*
  - Designed a framework for task planning in environments where agents' actions can have probabilistic effects, enabling planning under uncertainty in the presence of multiple agents.
  - My contribution: I mentored the team in the designing of the framework and helped conducting experiments for different ablations.
  - Accepted in ECMR 2025.

PUBLICATIONS

- Anticipate & Act: Integrating LLMs and Classical Planning for Efficient Task Execution in Household Environments: R Arora*, **Shivam Singh***, K Swaminathan, A Datta, S Banerjee, B Bhowmick, KM Jatavallabhula, M Sridharan, M Krishna.
  Accepted at IEEE International Conference on Robotics and Automation (ICRA) 2024 , Yokohama, Japan.
- AdaptBot: Combining LLM with Knowledge Graphs and Human Input for Generic-to-Specific Task Decomposition and Knowledge Refinement: **Shivam Singh**, K Swaminathan, N Dash, R Singh, S Banerjee, M Sridharan, M Krishna.
  Accepted at IEEE International Conference on Robotics and Automation (ICRA) 2025 , Atlanta, US.

- **Anticipate, Adapt, Act: A Hybrid Framework for Task Planning:** N Dash, A Kaura, **Shivam Singh**, R Singh, S Banerjee, M Sridharan, M Krishna.
  Accepted at IEEE European Conference on Mobile Robots (ECMR) 2025.

- **Pause and Think: A Dataset and Benchmark for Video-Grounded Assistive Action Suggestion:** **Shivam Singh**, S Majumder, P Prabhanjan, Z Liu, E Barsoum.
  arXiv preprint arXiv:2606.00616, 2026. Work done at AMD (Advanced Micro Devices, Inc.).

WORKSHOPS

- **DaTAPlan: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration:** **Shivam Singh***, K Swaminathan*, R Arora*, R Singh, A Datta, D Das, S Banerjee, M Sridharan, and M Krishna.
  Accepted in ICRA 2024 Workshop on Cooking Robotics: Perception and motion planning, *Yokohama, Japan*.

- **Anticipate & Collab: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration**: **Shivam Singh***, K Swaminathan*, R Arora*, D Das, S Banerjee, M Sridharan, M Krishna.
  Accepted in RSS2024 Workshop on Task Specification for General-Purpose Intelligent Robots, Delft, Netherlands

PROJECTS

- **Paper implemetation: A Point Set Generation Network for 3D Object Reconstruction from a Single Image** *2023*
  Pytorch, Python
  - Developed an architecture for generating 3D point clouds from single 2D images, outperforming previous methods in single-view 3D reconstruction.

- **Paper implemetation: *MonoLayout*: Amodal scene layout from a single image** *2024*
  Pytorch, Python
  - Implemented a deep learning model to predict road and vehicle layouts in bird's eye view from a single image.
  - Hallucinated occluded regions using adversarial learning for enhanced scene understanding.

- **Paper implemetation: Masked-attention Mask Transformer for Universal Image Segmentation** *2024*
  Pytorch, Python
  - Developed a universal image segmentation architecture capable of addressing any segmentation task (panoptic, instance, or semantic).
  - Utilized masked attention within the Transformer decoder to focus on localized features, significantly enhancing the model's performance on complex segmentation tasks.

SKILLS

Areas of interest: Machine Learning, Computer Vision, Vision-Language & Multimodal Models, Video Understanding, Robotics, Control Theory
Languages: Python, C++, C, PDDL, RDDL, HTML, RDF, SPARQL, Matlab
Technology: PyTorch, Hugging Face Transformers, LLaMA-Factory, vLLM, Keras, OpenCV, Open3D, Git, LaTeX
OS: Windows, Linux, ROS
Data Structures and Algorithms
Simulation Software: Gazebo, CoppeliaSim, AI2-THOR